

symbolicJacobian.m Documentation

`symbolicJacobian.m`

Purpose

Derives the robot geometric Jacobian **symbolically**:

$$J(q)$$

such that:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q}$$

Used for:

- Inspecting the Jacobian equations
- Verifying `controlJacobian.m`
- Singularity analysis
- Differential IK

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Main Method

The script uses the same Standard DH chain as FK.

For every revolute joint:

$$J_{v_i} = z_{i-1} \times (p_{TCP} - p_{i-1})$$

$$J_{\omega_i} = z_{i-1}$$

Therefore:

$$J_i = \begin{bmatrix} J_{v_i} \\ J_{\omega_i} \end{bmatrix}$$

and:

$$J = \begin{bmatrix} J_1 & J_2 & J_3 & J_4 & J_5 & J_6 \end{bmatrix}$$

⋮

Symbolic Jacobian

For compact notation define:

$$q_{23} = q_2 + q_3$$

$$q_{234} = q_2 + q_3 + q_4$$

Joint 1

$$J_1 = \begin{bmatrix} J_{v_1} \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

where:

$$J_{\omega_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and the linear part is:

$$J_{v_1} = \begin{bmatrix} J_{11} \\ J_{21} \\ 0 \end{bmatrix}$$

with:

$$J_{11} = \cos q_1 \left(\frac{2183}{20000} + \frac{823}{10000} \cos q_5 \right) + \sin q_1 \left(\frac{17}{40} \cos q_2 + \frac{1569}{4000} \cos q_{23} + \frac{823}{10000} \cos q_{234} \sin q_5 - \frac{1893}{20000} \sin q_{234} \right)$$

$$J_{21} = \sin q_1 \left(\frac{2183}{20000} + \frac{823}{10000} \cos q_5 \right) + \cos q_1 \left(-\frac{17}{40} \cos q_2 - \frac{1569}{4000} \cos q_{23} - \frac{823}{10000} \cos q_{234} \sin q_5 + \frac{1893}{20000} \sin q_{234} \right)$$

Joint 2

Define:

$$A_2 = \frac{1893}{20000} \cos q_{234} + \frac{1569}{4000} \sin q_{23} + \frac{17}{40} \sin q_2 + \frac{823}{10000} \sin q_{234} \sin q_5$$

Then:

$$J_2 = \begin{bmatrix} \cos q_1 A_2 \\ \sin q_1 A_2 \\ \frac{1893}{20000} \sin q_{234} - \frac{1569}{4000} \cos q_{23} - \frac{17}{40} \cos q_2 - \frac{823}{10000} \cos q_{234} \sin q_5 \\ \sin q_1 \\ -\cos q_1 \\ 0 \end{bmatrix}$$

$\mathcal{J}$

Joint 3

Define:

$$A_3 = \frac{1893}{20000} \cos q_{234} + \frac{1569}{4000} \sin q_{23} + \frac{823}{10000} \sin q_{234} \sin q_5$$

Then:

$$J_3 = \begin{bmatrix} \cos q_1 A_3 \\ \sin q_1 A_3 \\ \frac{1893}{20000} \sin q_{234} - \frac{1569}{4000} \cos q_{23} - \frac{823}{10000} \cos q_{234} \sin q_5 \\ \sin q_1 \\ -\cos q_1 \\ 0 \end{bmatrix}$$

$\mathcal{J}$

Joint 4

Define:

$$A_4 = \frac{1893}{20000} \cos q_{234} + \frac{823}{10000} \sin q_{234} \sin q_5$$

Then:

$$J_4 = \begin{bmatrix} \cos q_1 A_4 \\ \sin q_1 A_4 \\ \frac{1893}{20000} \sin q_{234} - \frac{823}{10000} \cos q_{234} \sin q_5 \\ \sin q_1 \\ -\cos q_1 \\ 0 \end{bmatrix}$$

$\mathcal{J}$

Joint 5

$$J_5 = \begin{bmatrix} -\frac{823}{10000} (\cos q_1 \cos q_5 \cos q_{234} + \sin q_1 \sin q_5) \\ \frac{823}{10000} (\cos q_1 \sin q_5 - \sin q_1 \cos q_5 \cos q_{234}) \\ -\frac{823}{10000} \sin q_{234} \cos q_5 \\ \cos q_1 \sin q_{234} \\ \sin q_1 \sin q_{234} \\ -\cos q_{234} \end{bmatrix}$$

$\mathcal{J}$

Joint 6

$$J_6 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \cos q_5 \sin q_1 - \cos q_{234} \cos q_1 \sin q_5 \\ -\cos q_1 \cos q_5 - \cos q_{234} \sin q_1 \sin q_5 \\ -\sin q_{234} \sin q_5 \end{bmatrix}$$

The first three entries are zero because, for the current UR5 model, the TCP lies on the Joint 6 rotation axis.

$\mathcal{J}$

Validation

The symbolic Jacobian is evaluated numerically using:

```
q_test = [  
    0.2  
   -0.3  
    0.5  
    0.1  
   -0.4  
    0.2  
];
```

and compared against:

```
controlJacobian(robot,q_test)
```

using:

```
symbolic_vs_numeric_difference = ...  
    J_symbolic_test - J_numeric_test;
```

and:

```
max_symbolic_error
```

The expected difference should be approximately machine precision.

Thus:

$$J_{symbolic} \approx J_{control}$$



Related Files

- [dhTransform.m](#) — Standard DH transformation
- [controlFK.m](#) — Numerical Forward Kinematics
- [symbolicFK.m](#) — Symbolic Forward Kinematics

- `controlJacobian.m` — Numerical geometric Jacobian